

Since you've already learned **AR** → **MA** → **ARMA** → **ARIMA** → **SARIMA** → **SARIMAX**, the next question is:

Why Do We Need VAR?

Problem with Previous Models

All previous models mainly forecast **one target variable**.

Example:

Model	Predicts
AR	One variable
MA	One variable
ARMA	One variable
ARIMA	One variable
SARIMA	One variable
SARIMAX	One target + external variables

Example in SARIMAX:

Weekly Sales

↓

Predicted using

Fuel Price

CPI

Unemployment

Only **Weekly Sales** is forecasted.

VAR Concept

VAR = **V**ector **A**uto**R**egression

In VAR:

ALL variables are forecasted together

Example:

Variable

Open

High

Low

Close

Volume

VAR assumes:

Today's Open
depends on
Yesterday's Open
Yesterday's High
Yesterday's Low
Yesterday's Close
Yesterday's Volume

Similarly:

Today's Close
depends on
Yesterday's Open
Yesterday's High
Yesterday's Low
Yesterday's Close
Yesterday's Volume

Real-Life Example

Reliance Stock

Suppose yesterday:

Variable	Value
Open	1400
High	1450
Low	1390
Close	1440
Volume	2 Million

Today:

Open
High
Low
Close
Volume

are all influenced by yesterday's values.

VAR captures this relationship.

SARIMAX vs VAR

SARIMAX	VAR
One target variable	Multiple target variables
External variables	All variables influence each other
Sales Forecasting	Multivariate Forecasting
Predict only Y	Predict all variables

Tamil Explanation

Imagine a family.

SARIMAX

Father's Salary

↓

Affected by

Fuel Price

Inflation

Only salary is predicted.

VAR

Father Salary

Mother Salary

House Expense

Savings

All affect each other.

Father Salary ↑

↓

Savings ↑

Savings ↑

↓

Expenses ↑

Expenses ↑

↓

Savings ↓

VAR models all these relationships simultaneously.

When to Use VAR?

Use VAR when:

✓ Multiple variables influence each other

Examples:

Domain	Variables
Stock Market	Open, High, Low, Close, Volume
Economy	GDP, Inflation, Interest Rate
Weather	Temperature, Humidity, Rainfall
Retail	Sales, Profit, Customers

Interview Answer

What is VAR?

VAR (Vector AutoRegression) is a multivariate time-series forecasting model that predicts multiple variables simultaneously by capturing the relationships among them. Each variable is modeled as a function of its own past values and the past values of all other variables in the system.

Why Learn VAR After SARIMAX?

Your learning path becomes:

AR
↓
MA
↓
ARMA
↓
ARIMA
↓
SARIMA
↓
SARIMAX
↓
VAR

At this stage, you move from **single-variable forecasting** to **multi-variable forecasting**, which is a major step toward advanced Time Series Analysis and Business Analytics.

For Your Portfolio

Reliance Industries Stock Forecasting Using VAR is an excellent next project because:

- Open, High, Low, Close, and Volume naturally influence each other.
- Easy to explain in interviews.
- Demonstrates multivariate forecasting skills.
- Complements your existing ARIMA/SARIMA/SARIMAX projects. 🚀

=====